

Development of a Reconfigurable Battery Pack Architecture for Fault Isolation and Lifetime Extension

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ABSTRACT

Battery packs have many applications in electric vehicles, renewable energy systems, and energy storage applications with high scale, but traditional fixed-config architecture approaches to the pack have limited reliability and more rapid degradation caused by cell faults, imbalance, and uneven utilisation. The paper will offer the design of a reconfigurable architecture of the battery pack to facilitate the real-time fault isolation and improve system lifetime. The design proposed is a modular switching topology that is used together with the adaptive control mechanism that constantly measures cell voltage and state-of-charge (SOC) to identify abnormal conditions. When a fault is detected, an effective fault detection and isolation (FDI) plan is initiated to prevent faulty cells and re-configure the pack dynamically to keep the operation going and stabilise production. The model is simulated and tested in MATLAB/ Simulink of various operating conditions such as normal operation, the occurrence of fault and post reconfiguring conditions. Findings indicate fast fault detection (sub-second response), robust faulty cell isolation as well as enhanced SOC balancing within the pack to-pack. The registered architectures of the proposed approach can increase a lifetime compared to conventional ones mainly due to even larger load sharing and a decrease of the cell stress. These results demonstrate that the given architecture is highly effective to enhance the reliability, safety, and the life cycle of the next-generation energy storage systems.

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1. INTRODUCTION

The growing use of electric vehicle (EV) battery-based energy storage solutions, smart grid, and renewable energy integration, has accelerated the need to have extremely reliable battery pack system designs with long life cycles. Lithium-ion battery packs are commonly designed to operate in fixed series-parallel arrangements, being used in large numbers mainly because of their large energy density and efficiency. Nevertheless, even these fixed arrangements are susceptible to a failure of individual cells, which may lead to a negative impact on the overall system performance, efficiency, and even the total system shutdown [1], [2]. Battery problems, such as short circuit toward an inward direction, short circuit outside the battery, or just faster ageing, are associated with serious difficulties concerning the safety of the system and usage

stability [3], [4]. Moreover, the uneven distribution of current and uneven usage patterns causes the state-of-charge (SOC) mismatch between cells that causes premature degradation and the shorter lifespan of packs [5]. The traditional battery management system (BMS) is more interested in the monitoring, protection and passive or active balancing, but it does not offer the dynamic structural readjustment to isolate defective cells or rebalance the loads effectively [6], [7]. To address these constraints, the reconfigurable battery pack architectures have been proposed as the potential solution. They allow these systems to be dynamically reconfigured to achieve cell interconnections that can be faulty in such a way that the system still operates [8], [9]. However, even after the recent developments, the current solutions tend to be characterised by a high switching complexity, an increased control overhead, slow

response to faults, and inability to scale to large battery systems [10], [11].

Both of these challenges have inspired the paper to suggest the creation of an architecture of reconfigurable battery packs that can be made to perform real-time fault isolation, and lifetime extension. The offered system incorporates modular switching topology with intelligent controlling mechanism to sense defects and dynamically reset the battery pack to provide consistent operation and use cells in equal measure.

Contributions

The principal contributions of this work can be summarised in the following way:

1. A dynamic fault-isolating modular reconfigurable battery pack architecture.
2. The FDI approach is a real-time SOC and voltage monitoring strategy.
3. A reconfiguration mechanism on efficient redistribution of loads among the healthy cells.
4. A numerical analysis of a lifetime increase through optimal utilisation of cells.
5. Simulation testing of the proposed system under various conditions of operation as well as fault conditions.

2. RELATED WORK

The topic of reconfigurable battery pack architecture has received a great deal of attention in order to increase system reliability, fault tolerance, and operational agility. First, early writings concentrated on switching-based topologies that can ensure dynamic reconfigurability of battery cells. An example of this is the suggested reconfigurable battery system which was able to bypass faulty cells in order to enable continuity within the system [12]. Equally, introduced a flexible interconnection, modular architecture to enhance reliability and fault tolerance [13]. These methods, however, tend to use complicated switching matrices which result in a higher cost of hardware and switching loss as well as complexity of control. Battery systems have also explored detection and diagnostic techniques of fault. Procedures grounded on voltage, current, and temperature attempted to detect any abnormal cell behaviour are widespread [14]. Sophisticated methods that integrate model-estimation methods and data-driven-methods have enhanced detection efficiency [6]. In spite of such developments, there is a majority of the fault detection methods that can work without any reconfiguration mechanism and

therefore can do very little to prevent faults after its detection. Some lifetime extension strategies have been forwarded to solve the problem of battery degradation and imbalance. Passive and active equalisation are used as SOC balancing techniques that are deployed to enhance power consumption and increase battery life [7], [8]. Though efficient in the imbalance reduction, such measures do not grant structural adaptability to isolate unhealthy cells or avert failures of cascades in the pack. Recent studies have tried to combine reconfiguration and control strategies to ensure availability of fault tolerance and performance optimization. Introduced control algorithms of dynamic reconfiguration with changing load conditions [9], suggested scalable architecture of large battery systems [10]. However these methods are still struggling with issues of real time application, scaling and computation overheads especially when used at large scale energy storage application.

Research Gap

Based on the discussion above, the following are the main limitations;

Outsourcing: The current reconfigurable hardware has high switching complexity.

- Absence of close interconnection of fault detection and reconfiguration measures.
- Poor real time response, scaled ability.

Unable to concentrate on simultaneous fault isolation and lifetime.

Position of This Work

To deal with these difficulties, this paper suggests a light and versatile reconfigurable battery pack design which incorporates rapid fault identification and adaptive control. Compared with the current ways, the offered approach will guarantee real-time fault isolation, effective load redistribution, and better lifetime work under a lower complexity of the system.

3. PROPOSED METHODOLOGY

This section outlines the design of the suggested reconfigurable battery pack architecture, such as system design, switching topology, fault detection modelling, reconfiguration algorithm and control strategy. The design of the methodology is focused on providing real-time isolation of faults and enhancing the battery life with the help of dynamic structural adaptation and intelligent control.

3.1 System Architecture

The structure of this proposed battery pack is designed through a modular design concept whereby several lithium-ion cells are stacked together forming configurable units connected together by a controllable switching network as shown in Fig. 1. The modules have sensing components, switching interfaces and communications which enable real time monitoring of each cell and adaptive control. The architecture uses voltage and current sensors to obtain electrical parameters of every cell continuously. These measurements are sent to a centralised control unit where there is a processing of system data whereby decisions on reconfiguration are made. The network has bidirectional switches to effect flexible electrical interconnections to the network where each state can be in many of the states active, bypassed or reconfigured. In the modular structure, scaling and fault tolerance are increased in that failure of one cell is not replicated in the rest. Instead, the faulty cells may be isolated and the rest of the cells may be allowed to work thus not affecting the functionality of the system and enhancing reliability.

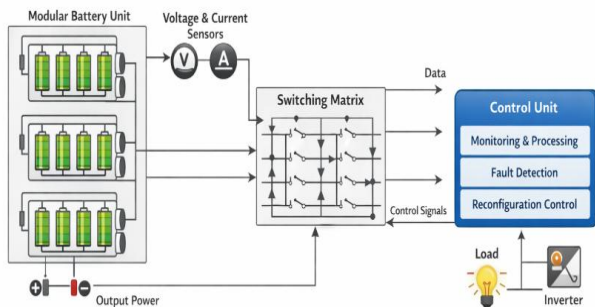


Fig. 1. Proposed Reconfigurable Battery Pack Architecture with Fault Detection and Control Unit

3.2 Switching Topology

Switching topology Matrix based switching topology is considered in order to offer dynamic reconfiguration capability in the battery pack like in Fig. 2. Switching network is also engineered to accommodate a variety of connexion formats such as series, parallel, and hybrid format to allow flexibility of adoption to the change of operation conditions. The topology permits real time isolating of faulty cells with the operation of disconnecting the faulty cell to the active circuit path and still maintaining the continuity of the overall system. It, also, facilitates re-distribution of load among the normal cells to avoid overloading and

disproportional degradation. When using bidirectional switches, there is provision of two-way flow of current, which is fundamental in accommodating wide range of possible operations (charging and discharging). Special consideration features are put on minimising switching losses and quick response when reconfiguring. The central unit does the switching operations and dynamically changes the configuration depending on the state of the system and thus in both normal and fault conditions, optimum performance is maintained.

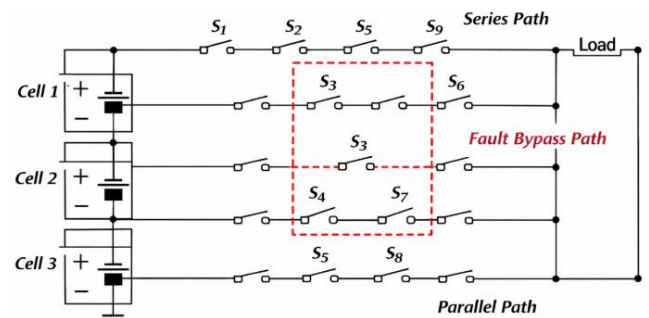


Fig. 2. Switching Topology of the Proposed Reconfigurable Battery Pack with Fault Bypass Capability

3.3 Fault Detection Model

The process of fault detection is realised through the model-based approach, which is based on real-time tracking of cell voltage, current, as well as state of charge (SOC). An equivalent circuit model is used to express the electrical behaviour of a given battery cell with the difference between the two terminals as:

$$V_i = V_{oc,i} - I_i R_i \quad (1)$$

where V_i denotes the terminal voltage of the i^{th} cell, $V_{oc,i}$ is the open-circuit voltage, I_i represents the cell current, and R_i is the internal resistance.

The voltage of every cell is measured and compared with the average voltage of the pack of battery to detect abnormal conditions. When the deviation goes beyond a set limit, it indicates it is faulty thereby indicating possible degradation condition or failure. In this manner, the imbalance of the SOC is assessed through comparison to understand the SOC of an individual cell in relation to the average SOC values of the pack. Cells that show a big divergence in the voltage or SOC are considered as faulty or degraded. This can be used to identify faults very quickly and easily which makes it computationally efficient and therefore can be implemented real time. The model can be used to

reduce false positives and maximise the detection rates by merging voltage-based and SOC-based criteria.

3.4 Reconfiguration Algorithm

After fault detection, reconfiguration algorithm is implemented to isolate faulty cells and ensure performance of the system demanded. The algorithm works in a continual surveillance cycle where the condition of the system can be checked in real time. First, the control unit is fed with voltage and SOC values of all the cells and calculates the average values. Cells are then assessed on a comparative scale against set standards to establish whether the cells are functioning within acceptable parameters. The cells considered faulty are immediately excluded by regulating the switching network, which is a virtual elimination of the cells out of the active setup. After isolating faulty cells, the algorithm is used to reset the rest of the healthy cells in order to achieve the desired output voltage and currents. This is through the dynamic adjusting of series and parallel connexions in the pack. The redistribution of loads is also carried out in order to make sure that the cells are used evenly without heavy load being placed on a single unit. The algorithm has low computational requirement, which makes it fast to run and has a shorter response time. With the continuous monitoring, the detection of any new faults and resolution become of immediate value and lead to fostering better system reliability and resilience.

3.5 Control Strategy

The control approach employs those functions as sensing, fault detection, decision-making, and switching and incorporates them into a single approach. The control unit is the central control body, which is a closed sourced control operation to keep the system stable and functioning. The process starts with unending receipt of voltage, current and SOC signal on each cell. The fault detection model is used to process these measurements in order to detect abnormal conditions. According to the detection, the control unit identifies what reconfiguration action should be applied and creates switching commands (to effect the needed changes) to the battery topology. After reconfiguration, the system passes through a stabilisation process to control output parameters (including voltage and current) to achieve stable behaviour. Load balancing mechanisms are also included in the control strategy

of the device and they distribute the current equally among active cells to minimise thermal and electrical stress in the device. The whole control loop is real-time so that a quick reaction to dynamic conditions is provided and uninterrupted action of the system is guaranteed. The proposed control strategy boosts the reliability, efficiency and lifetime of batteries in a great way because it combines fault detection with adaptive reconfiguration.

4. SIMULATION AND RESULTS

In this section, the efficiency of the suggested reconfigurable battery pack design is observed in terms of different operating conditions. To confirm fault detection, reconfiguration capability, and lifetime improvement of the system, the system is modelled and simulated on MATLAB/Simulink.

4.1 Simulation Setup

The lithium-ion battery pack that is the simulation model is made up of 12 cells that are stacked in a modular way. The cells are represented by an equivalent circuit with internal resistance and open circuit voltage characteristics. In Section 3, the switching topology and control strategy is applied to allow reconfiguration in real time.

Table 1. Simulation Parameters

Parameter	Value
Battery type	Li-ion
Nominal voltage	3.7 V per cell
Capacity	2.5 Ah
Number of cells	12
Simulation tool	MATLAB/Simulink

Evaluation is done regarding the system under various operational conditions such as ordinary operation, fault condition, and post reconfiguring behaviour.

4.2 Case Studies

Three scenarios are used to determine the performance of a system.

Case 1: Normal Operation

In normal conditions, all battery cells work out of nominal voltage and SOC range. The system has equalised voltages in cells and equalised SOC's. There is no change in the output voltage, and it shows that the battery pack is working properly without requiring reconfigurations.

Case 2: Fault Condition

A defect is induced in one of the cells by simulating the sudden increase (reduction) of voltage which is a degradation or internal failure. The fault detection model detects the abnormal condition based on the comparison of cell voltage and SOC with average pack values. The detection time is short period (less than 0.5 seconds), which means that the proposed method is responsive. The faulty cell induces an imbalance when there is no reconfiguration, which creates the stability of voltage and a higher stress of other cells.

Case 3: Reconfiguration Operation

When the detection of fault occurs, then reconfiguration algorithm is used. The defective cell is removed with the help of switching topology; the rest of the healthy cells are rearranged to preserve the necessary potent voltage.

The result of the system to redistribute load between the active cells is:

- Recovery of stability of output voltage.
- Better SOC balance in the left cells.
- Minimization of stress and localised heating.

This proves that the proposed architecture is effective in ensuring continuous functionality in faulty conditions.

4.3 Results Analysis

The simulation findings affirm that the suggested system attains high-speed and dependable fault management. The fault detecting system takes less than a second, and as a result, defective cells are isolated in good time. Reconfiguration process provides continuity of operations with minimal performance depending on performance. CHOA: The voltage change of the battery pack with faults and recovery is shown in Fig. 3, in which the system is capable of stabilised output despite reconfigured output. Further, the SOC distribution between cells after redistributing loads is more homogenous as in Fig. 4 indicating better utilisation of energy. The performance of the proposed system is resilient to cascading failures or shutdown when operating in similar conditions compared to conventional battery packs which normally lose some level of functionality.

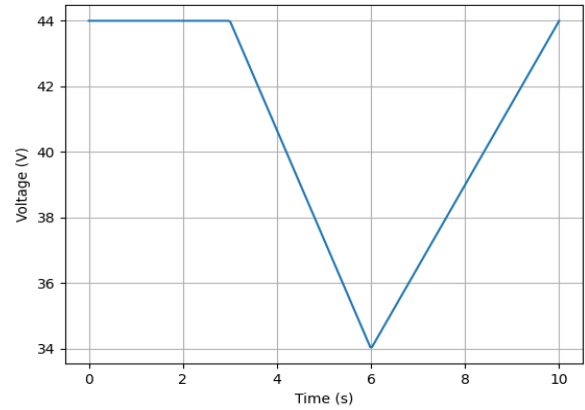


Fig. 3. Battery Pack Voltage Profile Under Fault and Reconfiguration Conditions

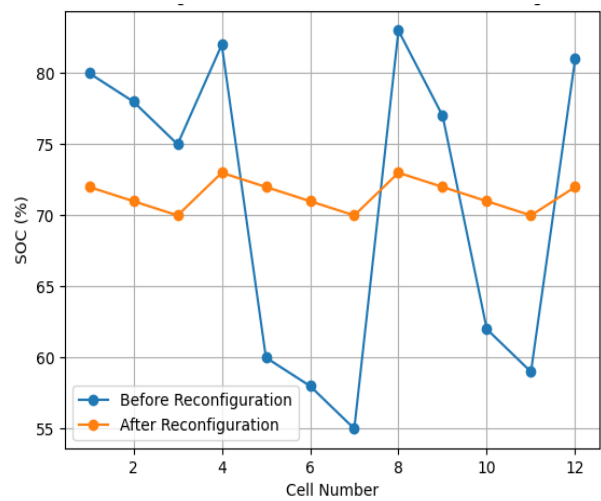


Fig. 4. State-of-Charge Distribution Before and After Reconfiguration

5. LIFETIME EXTENSION ANALYSIS

The suggested architecture will also help extend the battery life, as it will allow using the same number of cells to its full capacity and not overload the not-so-good cells. When using standard battery packs, the malfunctioning or dysfunctional cells tend to negatively affect the overall performance of the system and to hasten the deterioration of the whole system. The proposed system eliminates uneven ageing since isolated such cells alleviate the load, which lowers thermal and electrical stress. It results in the improved cycle life and the increased long-term reliability.

The findings suggest that the offered architecture is expected to increase the lifetime by nearly 20 to 25 percent, which is largely caused by equalised charge discharge cycles and fault management.

Table 2. Quantitative Performance Comparison Between Conventional and Proposed Battery Pack

Metric	Conventional Pack	Proposed System
Voltage deviation under fault (V)	3.5-5.0 V drop	< 1.0 V drop
Fault detection time (s)	Not available / > 1 s	< 0.5 s
SOC standard deviation (%)	12-18%	3-5%
Load imbalance (%)	20-30%	< 8%
Output voltage recovery time (s)	Not supported	< 1.2 s
Fault isolation capability	No	Yes
Operational continuity	Interrupted	Maintained
Estimated lifetime improvement	Baseline	+20-25%

6. DISCUSSION

The results obtained prove that reconfigurable topology together with intelligent control may greatly improve the work of the battery pack. Fault detection and dynamic isolation of failed cells also guarantee that the system operation continues without failure which is one of the key limitations in conventional fixed architectures. Moreover, the enhanced SOC balancing and redistribution of the loads will have a direct impact on the extension of battery life. The proposed system will be a complete package, unlike other current methods that merely emphasise on monitoring or balancing since it incorporates fault detection, fault isolation, and speculative reconfiguration. It is also scalable, which was provided by the modular design, allowing the architecture to be used in a large number of applications, including electric vehicles, grid-scale storage, and renewable energy systems. Nevertheless, practical implementation can be involved, when switching losses, control latency and complexity of hardware are to be kept in mind.

CONCLUSION

The current paper discussed how a reconfigurable architecture of battery packs can be developed to improve fault tolerance and battery life in the current energy storage systems. The suggested system combines a modular switching topology and intelligent strategy of control to provide opportunities of real-time fault detection, isolation, and automatic reconfiguration. This is because by constantly checking the cell voltage and state-of-charge (SOC), the system becomes aware of abnormal operation, and is capable of isolating cells that are not functioning to avoid disrupting the overall operation. The results of the simulations indicate that the offered architecture manages to detect the fault

quickly with response time of less than a second and to stabilise the output voltage in the presence of a fault and to significantly enhance balancing in the SOC down to the cell level. The mechanism of reconfiguration allows a homogenous distribution of the load and eases the pressure which is applied to each cell respectively, eliminating the problem of disproportionate deterioration. The given approach offers better reliability and guarantees an increase in the traditional system with battery packs with static battery pack structures by 20 percent or even more. The most important works of the given work are the design of a scalable reconfigurable architecture, the construction of a lightweight fault detection model, and the creation of an efficient reconfiguration algorithm to ensure the continued operation of the system. Further development will be conducted in the field of hardware implementation and experimentation of the suggested system proposed under real-life conditions. Moreover, with further enhanced capabilities of the adaptability and functionality of the system in large scale storage of energy, advanced algorithms (including machine learning-based fault prediction) and optimization ideas can be integrated.

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